(17)

$H_1: \sigma_1^2 = \sigma_2^2$; $H_a: \sigma_2^2 > \sigma_1^2$.

$F^* = \frac{S_2^2}{S_1^2} = \frac{28.2^2}{15.6^2} = 3.268$.

$F_{29,29} < F^*$ → Reject H_0 at $\alpha = 0.1$.

(16).

$H_0: \sigma_1^2 = \sigma_2^2$; $\sigma_1^2 \neq \sigma_2^2$.

$F^* = \frac{S_1^2}{S_2^2} = \frac{114^2}{103^2} = 1.22$.

$F_{14,14} \approx 2.12$.

$P(F > 1.22) > 0.1$

⇒ p-value $= 2P(F > 1.22)$

p-value > 0.2

⇒ Fail to reject H_0

(22)

a) $H_0: \sigma_1^2 = \sigma_2^2$; $H_a: \sigma_1^2 \neq \sigma_2^2$

$F^* = \frac{S_1^2}{S_2^2} = \frac{1.54}{2.96} = 0.52$.

p-value : $2 \times P(F_{14,14} > 0.52) > 0.2$

⇒ Fail to reject H_0.

b) 99% CI $= \left[\frac{S_1^2}{S_2^2} \times \frac{1}{F_{14,14,0.005}} ; \frac{S_1^2}{S_2^2} F_{14,14,0.005}\right]$

$= [0.122 ; 2.21]$

(11)

$H_0: \mu_1 = \mu_2$ $\quad \bar{X}_1 = 1.875$; $\bar{X}_2 = 2.625$.

$H_1: \mu_1 \neq \mu_2$ $\quad S_1 = 0.835$; $S_2 = 0.916$.

$S_1^2 = 0.6964$; $S_2^2 = 0.8393$.

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2} \approx 0.7679.$$

$$t^* = \frac{\bar{X}_1 - \bar{X}_2}{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \approx -1.71 \qquad |t| < 2.145.$$

$\Rightarrow$ Fail to reject H_0.

(12).

$H_0: \mu_1 = \mu_2$

$H_a: \mu_1 \neq \mu_2$.

$\bar{d} = -0.75$

$S_d = 0.886$.

$S_d^2 = 0.7857$.

$$t^* = \frac{\bar{d} - D}{S_d/\sqrt{n}} \approx -2.39.$$

$|t^*| > 2.365$.

$\Rightarrow$ Reject H_0.